Design and Implementation of a low-cost, open source IoT system using ESP32 with OLED, ThingsBoard and MQTT protocol

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Introduction

Interconnection of Renewable generation sources to generate and supply energy presents numerous challenges.

Diverse sensors, micro-controllers, micro-processor (MCU), actuators, valves, pumps, etc.

Supervisory Control and Data Acquisition (SCADA): telemetry and data acquisition

Data acquisition, Data presentation, Supervisory control, Networked data communication, Alarm processing, Historic data storage, data trending and reporting, and Remote monitoring.

Overview of Technologies

- Internet of Things (IoT)

Three/Five-layer architectures

MQTT(Application) – TCP (Transport Layer) – IPv4 (Internet Layer) – IEEE 802.11 (Network Access Layer)

- Message Queuing Telemetry Transport (MQTT) Protocol

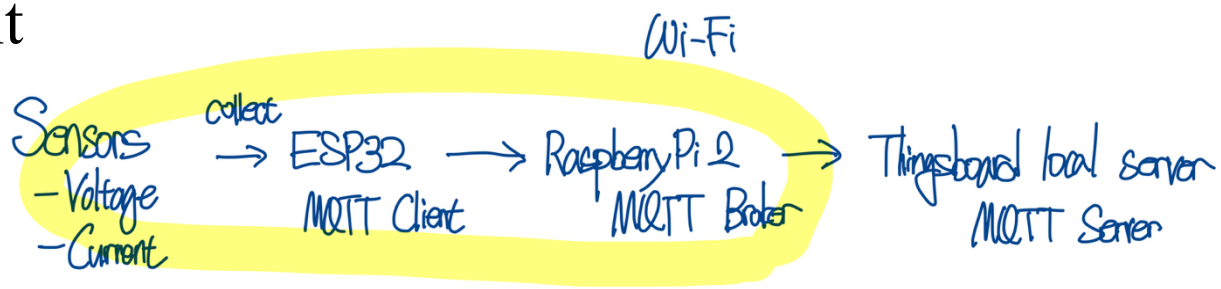
2 bytes fixed header -> low bandwidths, low computational power, low memory, low battery usage.

Broker can serve several Subscribers and Publisher

Quality of Service: three levels, reliable and suitable for IoT applications

SCADA System Components

- Sensors: MH Electronic Voltage Sensor modules (voltage sensor), ACS 712 Hall Effect Current Sensor (current sensor)
- TTGO ESP32 LoRa32 OLED micro-controller (RTU): MQTT Client, measure real-time PV voltage and current values, display OLED, publishes to MQTT broker
- Raspberry Pi single-board Computer: ThingsBoard IoT server (MQTT Broker)
- Wi-Fi Router (TCP/IP Wi-Fi connection): wireless connection between MQTT Client (ESP32) to MQTT Broker (Raspberry Pi)
- ThingsBoard local server IoT platform: Data collection, processing, visualization, and device management



Question:

I learned the importance of power in IoT. According to the paper, wireless routers consume a significant portion of total power. In reality, installing routers in certain IoT environments and establishing network communications can be challenging. What Solutions are available?

Answer:

My research online and through ChatGPT suggests that cellular networks are used to transmit data in remote areas without Wi-Fi. Low power cellular networks like LT-M and NB-IoT can be used to transmit network packets to a hub using base station signals.